G.C.E. (O/L) · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 04

Mathematics I

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Time: 2 hours **Total marks:** 100

Index number: .....

Important:

- Answer **all** questions on this question paper itself.
- Use only the space provided under each question for your working and your answer.
- Show the relevant steps and write the correct units in your answers.
- Marks: each question in Part A carries 2 marks; each question in Part B carries 10 marks.
- Scrap paper may be used for rough calculations. Use a ruler and a compass where needed.

Part	A (1–25)	B1	B2	B3	B4	B5	Total
Marks	/50	/10	/10	/10	/10	/10	/100

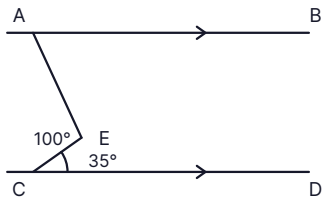
Part A

Answer all questions on this paper itself. Take $\pi = 22/7$ where needed. (2 marks each, 50 marks)

1. A crew of 12 workers can repair a section of railway track in 10 days. How many more workers, working at the same rate, are needed to complete the repair in 6 days?

Answer: .....

2. In the figure, AB ∥ CD, ∠AEC = 100° and ∠ECD = 35°. Find the magnitude of ∠BAE.



Answer: .....

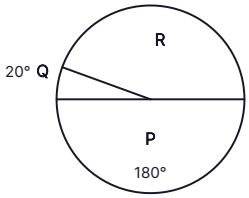
3. Simplify: $\frac{5}{2x} + \frac{7}{3x} - \frac{1}{4x}$

Answer: .....

4. Which one of the following quadrilaterals can **NOT** be a face of a right prism whose base is a triangle? Write its number.
(i) a square (ii) a rhombus that is not a square (iii) a rectangle

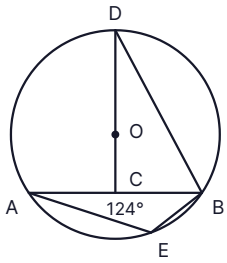
Answer: .....

5. The pie chart shows the quantities of three kinds of fish caught in a fishing village in one day: P – tuna, Q – seer fish, R – prawns. The quantity of seer fish is 30 kg. Find the quantity of prawns, in kg.



Answer: .....

6. In the figure, O is the centre of the circle and C is the midpoint of the chord AB. DOC is a straight line and ∠AEB = 124°. Find the magnitude of ∠CDB.



Answer: .....

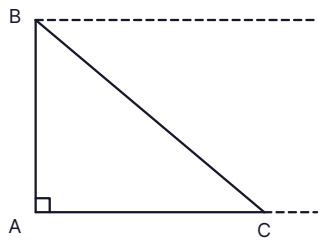
7. Factorise: $5x^2 - 45$

Answer: .....

8. Given that $10^{0.8451} = 7$, write down the value of lg 70.

Answer: .....

9. AB is a vertical lighthouse (A at the foot, B at the top) and a boat is at C, at the same horizontal level as A. In the figure, mark the angle of depression of C from B as a, and the angle of elevation of B from C as b. Write the relationship between a and b.



Answer: .....

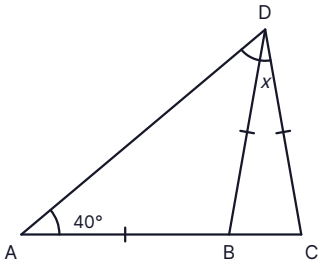
10. Chamari bakes a cake in a cylindrical tin whose height is 6 times its radius. Find the ratio of the area of paper needed to wrap the curved surface of the tin to the area of the paper needed to cover its base. (Curved surface area of a cylinder of radius r and height h is $2\pi rh$.)

Answer: .....

11. Two fair tetrahedral dice have their faces numbered 2, 3, 5 and 8. They are thrown together and the numbers on the bottom faces are added. Find the probability that the sum is 10.

Answer: .....

12. In the figure A, B and C lie on a straight line. AB = BD = DC and $\angle BAD = 40°$. Find the value of x.



Answer: .....

13. Let $\xi = \{x : x \text{ is a positive integer}, x \leq 10\}$, A = {prime numbers in ξ} and B = {odd numbers in ξ}. Write the number of the correct statement.
(i) $A \cup B = \{1, 3, 5, 7\}$ (ii) $A \cap B = \{3, 5, 7\}$ (iii) $n(A \cup B) = 7$

Answer: .....

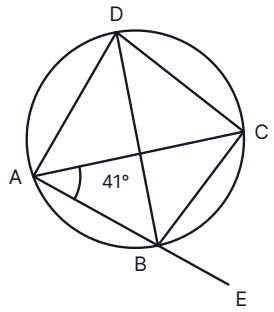
14. Simplify: $(6x^3y^2 \times 4xy^3) \div 8x^2y$

Answer: .....

15. The second term of a geometric progression is −6 and its third term is −18. Find the fifth term.

Answer: .....

16. In the figure, ABCD is a cyclic quadrilateral and BD bisects ∠ADC. AB is produced to E and ∠BAC = 41°. Find the magnitude of ∠CBE.



Answer: .....

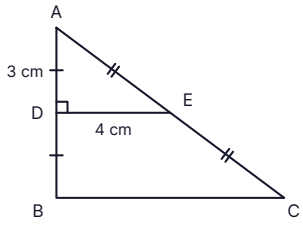
17. Find the highest common factor of $16x^2y$, $24xy^2$ and $40x^2y^3$.

Answer: .....

18. A bakery uses a triangular prism-shaped bread tin whose base has an area of 3850 cm² and whose height is 12 cm. A cylindrical tin of the same height holds the same volume. Find the radius r of the cylindrical tin. (Take π = 22/7; volume of a cylinder = $\pi r^2 h$.)

Answer: .....

19. In the triangle ABC in the figure, D and E are the midpoints of AB and AC respectively, ∠ADE = 90°, AD = 3 cm and DE = 4 cm. Find the perimeter of the trapezium BCED.

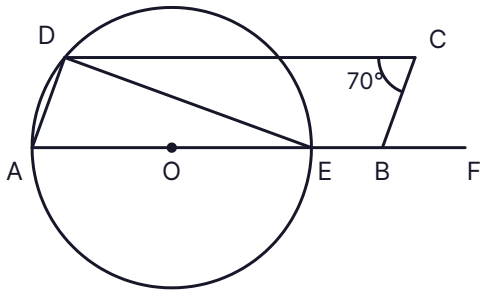


Answer: .....

20. Find the equation of the straight line passing through the points (2, 5) and (4, 11).

Answer: .....

21. In the figure, O is the centre of the circle through A, D and E. ABCD is a parallelogram and A, O, E, B, F lie on a straight line. ∠BCD = 70°. Find the magnitude of ∠DEF.

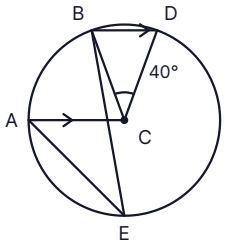


Answer: .....

22. Rashmi invested a sum of money for 2 years at an annual compound interest rate of 5%. The interest she earned in the second year was Rs. 2100 and the total amount she had at the end of the 2 years was Rs. 44 100. Find the sum she invested.

Answer: .....

23. In the figure, C is the centre of the circle, AC ∥ BD and ∠BCD = 40°. Find the magnitude of ∠AEB.

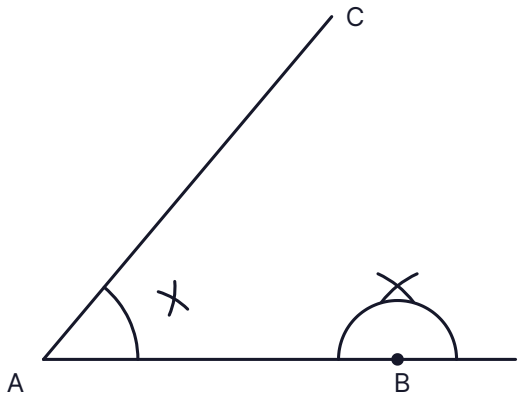


Answer: .....

24. Solve the inequality 4x + 23 > 2 and write the negative integer values of x that satisfy it.

Answer: .....

25. A circle touches the straight line AB at B and also touches the straight line AC. Some construction lines have been drawn in the figure. Complete the construction using them, and mark the centre O of the circle.



Answer: .....

Part B

Answer all questions on this paper itself. (10 marks each, 50 marks)

1. Lahiru, a railway engineer, pays income tax at 8% on the part of his monthly salary that exceeds Rs. 60 000. Of the money he receives after paying the tax, he spends $\frac{1}{8}$ on food and $\frac{2}{7}$ of the remainder on other expenses.

(i) What fraction of the money he receives remains after buying food? (1 mark)

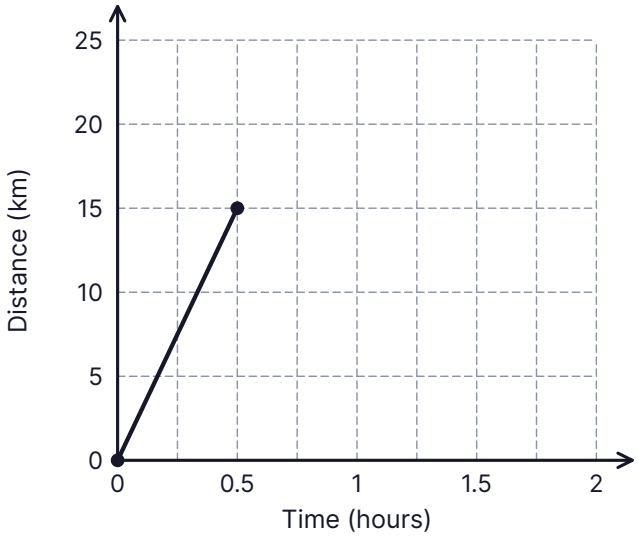
(ii) What fraction of the money he receives remains after spending on both food and other expenses? (2 marks)

(iii) The money that remains at the end is Rs. 72 000. Find the amount he spends on food and the amount he spends on other expenses. (2 marks)

(iv) Find his monthly salary before tax. (3 marks)

(v) Suppose he were exempt from tax and received his whole salary, but the amount spent on food stayed the same as in (iii). What percentage of his salary would the food expense be? (2 marks)

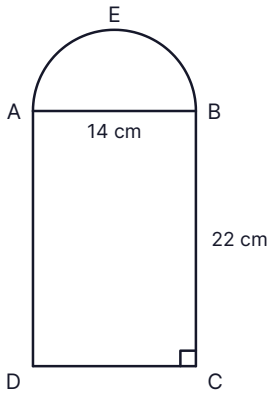
2. (a) Dinuka lives in a fishing village. He cycles from his home to the harbour at a uniform speed for 30 minutes and reaches it after covering 15 km. The first part of his distance–time graph is drawn on the grid below. He stays at the harbour for 30 minutes and then cycles back home along the same road at another uniform speed, taking 45 minutes.



(i) Complete the distance–time graph on the grid. (3 marks)

(ii) Find his speed on the return journey in km h^{-1}. (2 marks)

(b) The figure shows a plane figure made of a semicircle AEB joined to the rectangle ABCD along AB, where AB = 14 cm and BC = 22 cm. (Take $\pi = 22/7$.)



(i) Find the perimeter of the figure. (2 marks)

(ii) The area of the rectangle is how many times the area of the semicircle? (3 marks)

3. Chamari owns a shop building next to her bakery and rents it out for Rs. 15 000 per month, collecting the rent for the whole year in one payment. The municipal council has assessed the annual value of the building at Rs. 120 000 and charges annual rates of 15% of the assessed value. She spends 10% of the rent she collects on maintenance. She adds some more money to what remains and buys shares of Rs. 50 each. Each share pays an annual dividend of Rs. 3, and she receives a total dividend of Rs. 10 800 for the year.

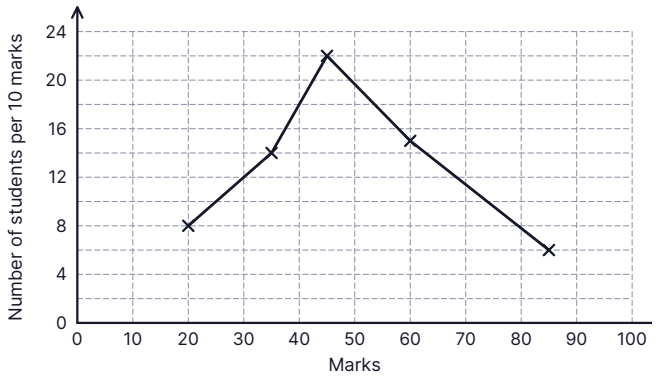
(i) Find the annual rates she has to pay. (2 marks)

(ii) Find the amount of money that remains after paying the rates and the maintenance cost from the annual rent. (2 marks)

(iii) (a) Find the number of shares she bought and the total amount she invested in them. (3 marks)

(b) Find the extra money she added, as a percentage of the total amount invested. (3 marks)

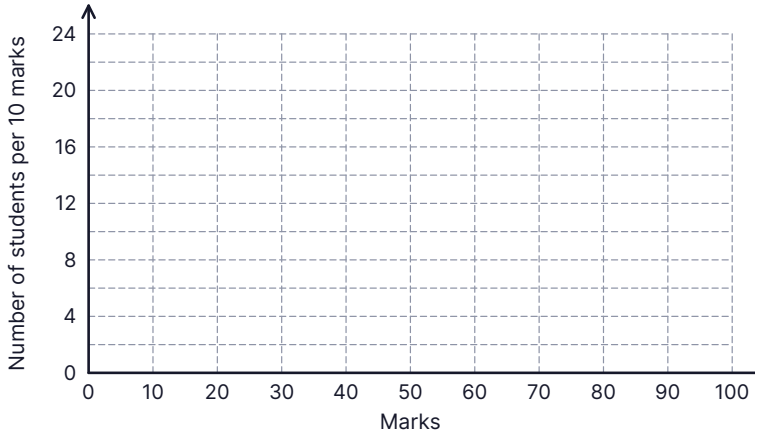
4. The marks scored by a group of students in a test are shown in the frequency polygon below. The heights of the plotted points (marked ×) at the class mid-values give the number of students per 10 marks in each class. The table of the data is incomplete.



Marks	Number of students	Cumulative frequency
10 – 30	16	16
30 – 40		
40 – 50		
50 – 70		
70 – 100		

(i) Complete the column of the number of students in the table. (The classes have unequal widths.) (3 marks)

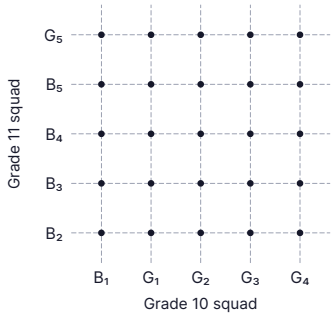
(ii) Draw the histogram for the data on the grid given. (3 marks)



(iii) Complete the cumulative frequency column of the table. (2 marks)

(iv) If a student is selected at random from the group, find the probability that the student scored 50 marks or more. (2 marks)

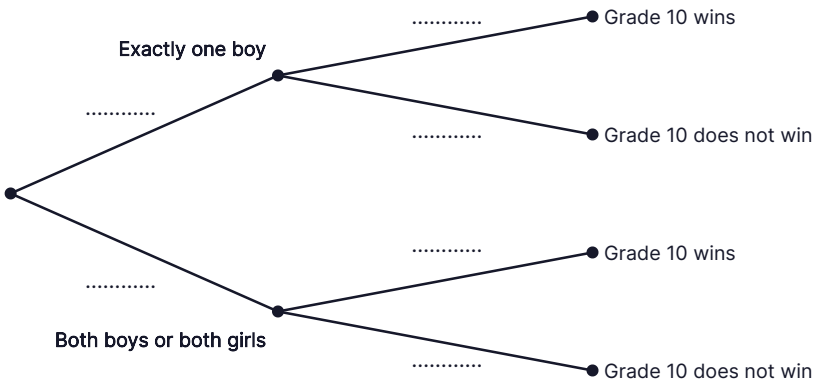
5. For a sports meet, the Grade 10 squad has one boy (B_1) and four girls (G_1, G_2, G_3, G_4), and the Grade 11 squad has four boys (B_2, B_3, B_4, B_5) and one girl (G_5). One member is chosen at random from each squad for the opening round. The 5 × 5 grid below is given, with the Grade 10 members across and the Grade 11 members up.



(i) On the grid, mark the sample space of the selection of the two members, using the symbol "×" for each outcome. (2 marks)

(ii) Enclose with closed curves the outcomes in which exactly one of the two chosen members is a boy, and find the probability of this event. (3 marks)

(iii) If exactly one of the two is a boy, the probability that the Grade 10 member wins the round is 1/5; if both are boys or both are girls, this probability is 3/4. Complete the probabilities on the tree diagram given. (3 marks)



(iv) Find the probability that the Grade 10 member does not win the round. (2 marks)

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